

Jiasheng (Jason) Hu

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I am a 3rd year PhD student in the department of Computer Science at University of Toronto, a member of Far Data Lab, supervised by Prof. Qizhen Zhang. My research interests mainly lie in cloud-native and distributed data processing systems, particularly their intersection with emerging hardware such as DPUs (also known as SmartNICs). The amount of data to be processed keeps growing, yet the CPU compute power is not keeping up; in short, I enjoy building and optimizing distributed data processing systems, with a focus on the network communication aspect.

EDUCATION

PhD Student in Computer Science, University of Toronto September 2023 – Present

3rd Year PhD Student @ Far Data Lab, supervised by Prof. Qizhen Zhang.

BEng in Computer Science, First Class Honours, University of Edinburgh September 2019 – May 2023

Honours Project: Homa transport protocol with zero-copy networking under Linux

RESEARCH & PUBLICATIONS

DPDPU: Data Processing with DPUs ([paper](#)) CIDR 2025

Jiasheng Hu, Philip A. Bernstein, Jialin Li, Qizhen Zhang

A vision paper where opportunities of utilizing DPUs to tackle existing problems in today's data processing systems are discussed, proposing the DPDPU framework, and addressing the related challenges and obstacles.

DDS: DPU-optimized Disaggregated Storage ([paper](#)) VLDB 2024

Qizhen Zhang, Philip Bernstein, Badrish Chandramouli, [Jason Hu](#), Yiming Zheng

One of the pioneering work in using DPU to accelerate Database systems, overhead and optimization opportunities within existing systems are identified, and we proposed partial offloading of read operations using DPU and the DDS framework, which resulted in an order of magnitude latency reduction, as well as up to tens of host CPU cores savings.

Combining Homa protocol with zero-copy networking ([code](#)) September 2022 - March 2023

Honours project, supervised by Prof. Michio Honda @ University of Edinburgh. Incorporating the Homa transport protocol in Linux Kernel, such as NVMe-oF and `sendpage` API; evaluating performance benefits of Homa for applications, such as gRPC and Redis.

ReFabric: Robotic system for automated textile sorting ([code](#)) January - May, 2022

System Design Project @ University of Edinburgh. Prizes won: Best Overall Project & Most Likely To Have A Positive Impact. Automated system for sorting textiles according to colors and materials, using machine learning models with camera images and infrared sensors.

TEACHING EXPERIENCES

Teaching Assistant at University of Toronto May 2024 - present

- Introduction to Databases (CSC343, CSCC43), Introduction to Computer Programming (CSC108), Mathematical Expression and Reasoning for Computer Science (CSC165)

Teaching Assistant and Volunteer Mentor at University of Edinburgh September 2021 - April 2023

- Introduction to Computation, System Design Project

VOLUNTEERING AND SERVICES

- Game Dev Society @ University of Edinburgh: organizing game jam events, industry talks etc.
- Tutor of Programming Club @ University of Edinburgh: teaching introductory computer programming, participating in various programming contests.